Latency Minimization for Uplink RSMA System

Shanik Hubert, *Student Member, IEEE*, and Carrson C. Fung*, *Member, IEEE*

Abstract

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A. Uplink Blocklength Minimization for Reduction of Asynchronicity

For MIMO uplink (UL), the receive signal is written as

$$\mathbf{y} = \sum_{k=1}^{K} \mathbf{H}_k \mathbf{F}_k \mathbf{s}_k + \mathbf{w} \in \mathbb{C}^{N_R},$$

where $\mathbf{H}_k \in \mathbb{C}^{N_R \times N_k}$ denotes the uplink channel between the kth user and the AP, $\mathbf{F}_k \in \mathbb{C}^{N_k \times d_k}$ denotes the precoding matrix for the kth user, d_k denotes the number of spatial streams user k is sending to the AP (and $d_k \leq N_k$), $\mathbf{s}_k \in \mathbb{C}^{d_k}$ denotes the transmitted signal for the kth user, $\mathbf{w} \in \mathbb{C}^{N_R}$ denotes the AWGN, with N_k denoting the number of antennas at the kth user and N_R denotes the number of antennas at the AP. It is assumed that $E\left[\mathbf{s}_k \mathbf{s}_k^H\right] = \mathbf{I}_{N_k}$. Let

$$x_k = \mathbf{F}_k \mathbf{s}_k,$$

so

$$\begin{aligned} E\left[|\mathbf{x}_k\|_2^2\right] &= \mathbf{F}_k E\left[\mathbf{s}_k \mathbf{s}_k^H\right] \mathbf{F}_k^H \\ &= \mathbf{F}_k \mathbf{F}_k^H \\ &= \|\mathbf{F}_k\|_F^2 \leq P_k, \end{aligned}$$

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Shanik Hubert and Carrson C. Fung are both affiliated with the Department of Electrical and Computer Engineering, National Yang Ming Chiao Tung University, Hsinchu 300, Taiwan. (e-mail: dmytro.ivakhnenkov@gmail.com, c.fung@ieee.org)

where P_k is the transmit power from the kth user. If SIC is used at the receiver, then the achievable rate at the receiver is

$$R_{sum} = \sum_{k=1}^{K} \log_2 \det \left(\mathbf{I}_{N_R} + \mathbf{H}_k \mathbf{F}_k \left(\sigma_w^2 \mathbf{I}_{N_R} + \sum_{j \neq k} \mathbf{H}_j \mathbf{F}_j \mathbf{F}_j^H \mathbf{H}_j^H \right)^{-1} \mathbf{F}_k^H \mathbf{H}_k^H \right)$$

Then it seems the latency minimization problem can be formulated as

$$\begin{aligned} \min_{\mathbf{F}_k, B_k, \forall k} \ & \sum_{k=1}^{K} \frac{B_k}{R_{sum}} \text{ (I think } R_k \\ s.t. \ & \|\mathbf{F}_k\|_F^2 \leq P_k, \forall k, \\ & B_k \in \{B_{k,\min}, \ldots, B_{k,\max}\}, \end{aligned} \tag{1}$$

where R_k is the Shannon rate for the kth user. (1) makes sense when infinite size blocklength is assumed. However, this assumption is usually incorrect when latency is minimized as the design variable should be the number of blocks (i.e. channel used) for transmission, implying the blocklength is short and finite. Possible extensions for (1) include optimizing not with the sum rate, but using rate such as

1. Max-min rate:

$$\max_{\mathbf{F}_k, \forall k} \min_k R_k.$$

   This has a strong fairness guarantee, often leads to equal rates across all users if we can find a feasible solution. This is the worst-case that Carrson was talking about.

2. The proportional fairness:

$$\max_{\mathbf{F}_k, \forall k} \sum_{k=1}^{K} \log(\mathbf{R}_k),$$

   which balances the throughput and fairness.

3. Weighted fairness

$$\max_{\mathbf{F}_k, w_k, \forall k} w_k \log(R_k),$$

   where w_k are weights for QoS differentiation, i.e., higher w_k is, more resource will be allocated the kth user.

B. Uplink Latency Minimization Using RSMA

Using RSMA for uplink transmission can theoretically acheive the optimal rate region according to [1] by splitting the original K input channels to $2K-1$ inputs (though the Kth user message need not be split, but most papers consider splitting the last user message for simplicity). The transmitted message of the kth user $\mathbf{x}_k$ is split into

$$\mathbf{x}_k = \sum_{i=1}^{2} F_{ki} \mathbf{s}_{ki},$$

where i denotes the stream index. So the received signal at the AP can be written as

$$\mathbf{y} = \sum_{k=1}^{K} \sum_{i=1}^{2} \mathbf{H}_{ki}\mathbf{F}_{ki}\mathbf{s}_{ki} + \mathbf{w} \in \mathbb{C}^{N_R},$$

where

$$\sum_{i=1}^{2} \|\mathbf{F}_{ki}\|_F^2 \leq P_k$$

SIC is assumed to be used to decode each user's signal at the AP, which involves the use of a decoding order π, let the permutation π belong to the set Π defined as the set of all possible decoding order of all $2K$ messages from K users. Let the decoding order of message s_{ki} from the kth user to be π_{ki}, then the achievable rate of decoded message s_{ki} can be given as

missing equation here.

Let $\mathcal{K}$ be the set of user devices, Let $\mathcal{J} = \{1, 2\}$ be the set of containing the number of split streams. Then the rate of the ith stream for user k is

$$R_{ki} = \log_2 \det \left(\mathbf{I}_{N_R} + \mathbf{H}_k \mathbf{F}_{ki} \left(\sigma_w^2 \mathbf{I}_{N_R} + \sum_{\{(j \in \mathcal{K}, i \in \mathcal{J}) | \pi_{ji} > \pi_{ki}\}} \mathbf{H}_j \mathbf{F}_{ji} \mathbf{F}_{ji}^H \mathbf{H}_j^H \right)^{-1} \mathbf{F}_{ki}^H \mathbf{H}_k^H \right),$$

where $\{(j \in \mathcal{K}, i \in \mathcal{J}) | \pi_{ji} > \pi_{ki}\}$ is the set that represents the messages s_{ji} that are decoded after message s_{ki}. Hence the total rate for user k is

$$R_k = \sum_{i=1}^{2} R_{ki}$$

To optimize the rate in uplink RSMA, a joint optimization of both the power management and message decoding order must be done, which results in solving

$$\begin{aligned} &\max_{\mathbf{F}_k, \forall k} \min_{k, \pi} R_k. \\ &\text{s.t.} \sum_{i=1}^{2} \|\mathbf{F}_{ki}\|_F^2 \leq P_k \\ &\pi \in \Pi, \|\mathbf{F}_{ki}\|_F^2 \geq 0, \forall k \in \mathcal{K}, j \in \mathcal{J} \end{aligned}$$

This is similar to the optimization problem modeled in [2] where they optimize for the sum rate.

C. Uplink Latency Minimization Using Finite Blocklength Minimzation

To consider latency in URRLC systems with finite blocklength, let us model channel rate not in terms of (bits/s) but as (bits/channel use). Let blocklength of the kth user be n_k or the number of channel used per symbol. Let T_s be the time taken to transmit one symbol. Then the latency of the kth user can be expressed as

$$\begin{aligned}\text{Latency}_k &= n_k \times T_s \\ &= \frac{B(\text{bits})}{R(\text{bits/channel use})} \times T_s(\text{sec})\end{aligned}$$

Thus latency for the kth user is optimized by optimizing for n_k.

According to [3], the **rate under finite blocklength** is expressed as

$$R^*(n,\epsilon) = C - \sqrt{\frac{V}{n}}Q^{-1}(\epsilon) + \mathcal{O}\left(\frac{\log n}{n}\right), \tag{2}$$

where C is Shannon capacity of channel, $Q^{-1}(\cdot)$ is the inverse of the Gaussian Q function, V is the channel dispersion. According to [3], (2) is used to sustain the desired error probability ϵ for a given packet size n, one incurs a penalty on the rate (compared to the channel capacity) that is proportional to $\frac{1}{\sqrt{n}}$.

According to [4], the **rate under finite blocklength in massive MIMO systems** in high SNR regime is approximated as

$$R_k \approx d_k \log\left(1+\rho_k\right) - \sqrt{\frac{d_k}{n_k}}Q^{-1}\left(\varepsilon_k\right).$$

which is the rate of user k. ρ_k is the SNR of user k and $Q^{-1}(\cdot)$ denotes the inverse of Gaussian Q function. At this point, we are not sure why [4] removed the channel dispersion V, which is part of (2). The inverse of the Q function is

$$Q^{-1}(x) \triangleq \int_x^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$

Rearranging the equation, when d_k is large enough,

$$n_k \approx \frac{d_k\left[Q^{-1}\left(\varepsilon_k\right)\right]^2}{\left[d_k \log_2\left(1+\rho_k\right) - R_k\right]^2} \propto \frac{1}{d_k},$$

This shows that the latency in the time domain can be reduced by increasing the DoFs in the space domain, for the purpose of maintaining a certain level of performance target. According to [4]

$$\begin{aligned}n_k &= B_k/R_k \\ &= \frac{B_k}{d_k \log\left(1+\rho_k\right) - \sqrt{\frac{d_k}{n_k}}Q^{-1}\left(\varepsilon_k\right)}.\end{aligned} \tag{3}$$

Solving (3) would give another equation that describe n_k if we know the number of bits B_k. We am not sure how this fits in with the other formulas. From (3), it can also be seen there are several ways to decrease n_k, which are

1) Increase spatial DoF d_k
2) Increase SNR ρ_k
3) Increase error probability ϵ_k (reliability)

The **min-max blocklength/Latency** can be expressed as

$$\min_{\forall k} \max_{k, R_k, \epsilon_k,} n_k.$$

D. Uplink Latency Minimization with RSMA and Finite Blocklength Minimization

[5] considered minimizing the blocklength in uplink RSMA in a simple 2-user system. However in our problem, we want to improve the latency of the worst-case user, i.e. the user with the worst channel (e.g. worst channel-to-noise (CNR)), we should consider increasing d_k, ρ_k, or ϵ_k of the weakest user through precoder design, while decreasing the same parameters of the stronger users so that all users will experience the same latency, thus reducing asynchronality among the users. Similarly reducing the error probability of the stronger user will be helpful as the data from teh stronger users are weighted higher. This helps to reduce the unfariness of asynchrononicity and in the context of federated learning, this ensures that the weaker users still have an impact on the training while not weighting the data higher due to higher error rate fro these users due to the poor channel. This is something that i do not believe is explored in [5].

E. Problem Formulation of Blocklength Reduction to Reduce Asynchronality

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The Shannon rate of the receiver can be defined as,

$$R_k = \log_2 \det \left(\mathbf{I}_{N_R} + \mathbf{H}_k \mathbf{F}_k \left(\sigma_w^2 \mathbf{I}_{N_R} + \sum_{j \neq k} \mathbf{H}_j \mathbf{F}_j \mathbf{F}_j^H \mathbf{H}_j^H \right)^{-1} \mathbf{F}_k^H \mathbf{H}_k^H \right)$$

Defining the latency of the kth use as

$$T_k = \frac{n_k}{f_s},$$

where T_k is the latency, $n_k \in \mathbb{Z}_+$ is the number of channel used and f_s is the symbol rate. Considering error probability ϵ and using the it to calculate latency under the mean no. of retranmsmissions

$$\mathrm{P}(n \text{ transmissions}) = \epsilon_k^{n-1}(1-\epsilon_k).$$
$$\mathbb{E}(\text{no. of transmissions}) = n_{mean} = \frac{1}{1-\epsilon_k}.$$

Sincer a finite blocklength n_k is considered, the upper bound to the rate of system for the kth user is written as [3]

$$\frac{B_k}{n_k} \leq R_k^*(n,\epsilon) = R_k - \sqrt{\frac{V_k}{n_k}}Q^{-1}(\epsilon_k),$$

where B_k is the fixed number of bits that need to sent by the kth user and n_k is the number of blocklength or channel uses or symbols for the kth user. Thus $R_k^*(n,\epsilon)$ can be defined in the unit of (bits/channel use). Then the weighted latency minimization problem can be formulated as

$$\begin{aligned} \min_{\mathbf{F}_k, n_k, \epsilon_k} \quad & \sum_{k\in K} W_k \frac{n_k}{f_s} \\ \text{s.t.} \quad & \frac{B_k}{n_k} \leq R_k(\mathbf{F}_k) - \sqrt{\frac{V_k(\mathbf{F}_k)}{n_k}}Q^{-1}(\epsilon_k), \quad \forall k \in \mathcal{K} \\ & n_k \in [n_{\min}, n_{\max}], \quad \forall k \in \mathcal{K} \\ & \|\mathbf{F}_k\|_F^2 \leq P_k, \quad \forall k \in \mathcal{K}. \end{aligned} \tag{4}$$

A more complete formulation by considering user specific ϵ_k and retransmission affected latency can be considered as

$$\begin{aligned} \min_{\mathbf{F}_k, n_k, \epsilon_k} \quad & \sum_{k\in K} W_k \frac{n_k}{f_s} \frac{1}{1-\epsilon_k} \\ \text{s.t.} \quad & \frac{B_k}{n_k} \leq (1-\epsilon_k)\left[R_k(\mathbf{F}_k) - \sqrt{\frac{V_k(\mathbf{F}_k)}{n_k}}\,Q^{-1}(\epsilon_k)\right], \quad \forall k \in \mathcal{K}, \\ & n_k \in [n_{\min}, n_{\max}], \quad \forall k \in \mathcal{K}, \\ & \epsilon_k \in [0, \epsilon_0], \quad \forall k \in \mathcal{K}, \\ & \|\mathbf{F}_k\|_F^2 \leq P_k, \quad \forall k \in \mathcal{K}. \end{aligned} \tag{5}$$

F. Strategies to solve the Problem Formulation

Solution for (4) can be attained via alternating optimization. First, by fixing $\mathbf{F}_k$, n_k can be found via exhaustive search by searching the interval $[n_{\min}, n_{\max}]$ for $k \in \mathcal{K}$. Then after a suboptimal (because n_k, $\forall k$ is found based on a suboptimal $\mathbf{F}_k$), then fixing $\mathbf{n}_k$, we should find $\mathbf{F}_k$, for $k \in \mathcal{K}$, by solving

$$\begin{aligned} &\max_{\mathbf{F}_k, k\in\mathcal{K}} R_k(\mathbf{F}_k) - \sqrt{\frac{V_k(\mathbf{F}_k)}{n_k}} Q^{-1}(\epsilon_k), \quad \forall k \in \mathcal{K} \\ &\|\mathbf{F}_k\|_F^2 \leq P_k, \quad \forall k \in \mathcal{K}. \end{aligned} \tag{6}$$

This implies if the problem needs to be convex, then $R_k(\mathbf{F}_k)$ needs to be concave (where the maximization of $R_k(\mathbf{F}_k)$ can be done by reformulating the problem into minimization of the WMMSE) and $\sqrt{\frac{V_k(\mathbf{F}_k)}{n_k}}$ needs to be convex. Since square root is a monotonically nondecreasing concave function, the square root can be removed and $V_k(\mathbf{F}_k)$ can be approximated, e.g. using its first-order Taylor series expansion, so that (6) can be convexified as

$$\begin{aligned} &\max_{\mathbf{F}_k, k\in\mathcal{K}} W_k(\mathbf{F}_k) - \left(V_k\left(\mathbf{F}_k^{(p)}\right) + \nabla V_k\left(\mathbf{F}_k^{(p)}\right)^H \left(\mathbf{F}_k - \mathbf{F}_k^{(p)}\right)\right), \quad \forall k \in \mathcal{K} \\ &\|\mathbf{F}_k\|_F^2 \leq P_k, \quad \forall k \in \mathcal{K}, \end{aligned} \tag{7}$$

where $\mathbf{F}_k^{(p)}$ is the precoder at the pth iteration of the successive convex optimization (SCA) (or the WMMSE update since the SCA needs to be applied in conjunction with the WMMSE iteration).

Since the rate of the kth user is defined as

$$R_k = \log\det\left(\mathbf{I}_{N_R} + \mathbf{H}_k\mathbf{F}_k\mathbf{F}_k^H\mathbf{H}_k^H\left(\sigma_w^2\mathbf{I}_{N_R} + \sum_{j\neq k}\mathbf{H}_j\mathbf{F}_j\mathbf{F}_j^H\mathbf{H}_j^H\right)^{-1}\right),$$

let $\mathbf{H}_k \in \mathbb{C}^{N_r\times N_k}$ with N_k denoting the number of transmit antennas at user k. This implies $\mathbf{F}_k \in \mathbb{C}^{N_k\times d_k}$, where $d_k \leq N_k$ denotes the number of spatial streams from the kth user. Then it has been shown in [6] that the weighted sum rate

$$\sum_{k=1}^{K} w_k R_k, \quad s.t. \, \|\mathbf{F}_k\|_F^2 \leq P_k$$

can be maximized by solving the WMMSE problem. Define the kth receiver at the AP as $\mathbf{U}_K \in \mathbb{C}^{N_R\times d_k}$. Also define the weight matrix $\mathbf{W}_k \in \mathbb{C}^{d_k\times d_k}$ where $\mathbf{W}_k \succ \mathbf{0}_{d_k\times d_k}$ (which will be equal to $\mathbf{E}_k^{-1}$). Further define the MSE matrix

$$\mathbf{E}_k \triangleq \mathbf{I}_{d_k} - \mathbf{U}_k^H\mathbf{H}_k\mathbf{F}_k - \mathbf{F}_k^H\mathbf{H}_k^H\mathbf{U}_k + \mathbf{U}_k^H\left(\sigma_w^2\mathbf{I}_{N_r} + \sum_{j=1}^{K}\mathbf{H}_j\mathbf{F}_j\mathbf{F}_j^H\mathbf{H}_j^H\right)\mathbf{U}_k \in \mathbb{C}^{d_k\times d_k}.$$

Then the per user matrix quadratic transform identity implies

$$\begin{aligned} &w_k\log\det\left(\mathbf{I}_{N_R} + \mathbf{H}_k\mathbf{F}_k\mathbf{F}_k^H\mathbf{H}_k^H\left(\sigma_w^2\mathbf{I}_{N_R} + \sum_{j\neq k}\mathbf{H}_j\mathbf{F}_j\mathbf{F}_j^H\mathbf{H}_j^H\right)^{-1}\right) \\ &= \max_{\mathbf{U}_k, \mathbf{W}_k\succ\mathbf{0}_{d_k\times d_k}} w_k\left(\log\det\mathbf{W}_k - tr\left(\mathbf{W}_k\mathbf{E}_k\right)\right) + \text{constant}. \end{aligned}$$

Then the original weighted sum rate prolbme $\max_{\mathbf{F}_k, k\in\mathcal{K}} w_k R_k$, subject to $\|\mathbf{F}_k\|_F^2 \le P_k$ problem is equivalent to

$$\begin{aligned} &\max_{\mathbf{F}_k, \mathbf{W}_k, \mathbf{U}_k, k\in\mathcal{K}} w_k \left(\log\det \mathbf{W}_k - tr\left(\mathbf{W}_k \mathbf{E}_k\right)\right) \\ &\quad s.t. \ \|\mathbf{F}_k\|_F^2 \le P_k. \end{aligned} \tag{8}$$

Let

$$\mathbf{\Sigma} \triangleq \sigma_w^2 \mathbf{I}_{N_R} + \sum_{j=1}^{K} \mathbf{H}_j \mathbf{F}_j \mathbf{F}_j^H \mathbf{H}_j^H.$$

Then the following WMMSE algorithm can be used to find the solution for (8).

1. **MMSE receiver update**

$$\mathbf{U}_k = \mathbf{\Sigma}^{-1} \mathbf{H}_k \mathbf{F}_k.$$

2. **Weight matrix computation (inverse MSE)**

$$\mathbf{W}_k = \mathbf{E}_k^{-1}$$

3. **Precoder computation (quadratic KKT with power constraint)** Let

$$\begin{aligned} \mathbf{C}_k &\triangleq \mathbf{H}_k^H \mathbf{U}_k \mathbf{W}_k \mathbf{U}_k^H \mathbf{H}_k \in \mathbb{C}^{N_k \times N_k} \\ \mathbf{D}_k &\triangleq \mathbf{H}_k^H \mathbf{U}_k \mathbf{W}_k \in \mathbb{C}^{N_k \times d_k}. \end{aligned}$$

Then the precoder with per-user power constraint is computed as

$$\mathbf{F}_k(\lambda_k) = \left(\mathbf{w}_k \mathbf{C}_k + \lambda_k \mathbf{I}_{N_k}\right)^{-1} \left(\mathbf{w}_k \mathbf{D}_k\right) \in \mathbb{C}^{N_k \times d_k},$$

where $\lambda_k \ge 0$ is chosen by the bisection method to satisfy $tr\left(\mathbf{F}_k \mathbf{F}_k^H\right) = P_k$ if needed, otherwise $\lambda_k = 0$ and $tr\left(\mathbf{F}_k \mathbf{F}_k^H\right) < P_k$. That is, this corresponds to the complementary slackness condition in the KKT conditions.

Note that it has been reported that by initializing $\mathbf{F}_k$ using MRT precoder speeds up the convergence of the WMMSE algorithm. Note also that the sum rate is maximized by maximizing the per-user rate R_k. Note that we may be able to set the weight w_k as the inverse of the CNR so that the rate of the worst-case (i.e. worst CNR) will be optimized the most.

Algorithm 1 summarizes the steps.

Algorithm 1: Proposed latency minimization algorithm in the uplink via precoder.

Result: $n_k, \ \mathbf{F}_k^{\star}, \forall k$

Initialization: Set $i = 0$, choose ε (e.g. small integer number), choose $\left\{\mathbf{F}_k^{(0)}, \forall k\right\}$ as the MRT precoder, w_k = inverse of the CNR$_k$

do

  Exhaustively find n_k, $\forall k$ using $\mathbf{F}_k^{(i)}$;

  Denote (sub-)optimal blocklength as $n_k^{\star}$;

  $n_k^{(i)} = n_k^{\star}$, $\forall k$;

  for *each user* $k = 1, \dots, K$ **do**

    Use the WMMSE algorithm and convex approximation of $V_k(\mathbf{F}_k)$ to find the precoder $\mathbf{F}_k$ using $n_k^{(i)}$;

    Denote (sub-)optimal precoder as $\mathbf{F}_k^{\star}$;

  end

  $i = i + 1$;

  $\mathbf{F}_k^{(i)} = \mathbf{F}_k^{\star}$, $\forall k$;

while $\left|\sum_{k\in\mathcal{K}} n_k^{(i)} - \sum_{k\in\mathcal{K}} n_k^{(i-1)}\right| \leq \varepsilon$;

G. Capacity and Channel Dispersion

The Capacity and Channel Variance of Coherant Rayleigh Fading Channel is discussed in [7].

However [7] makes the folowing assumptions are made,

1) The channel is isotrophic.
2) The noise has unit variance.
3) Considers SU-MIMO.

The Capacity of the channel of user-k is defined as,

$$\begin{aligned} C(P) &= \frac{1}{2}\mathbb{E}\left[\log\det\left(I_{n_r} + \frac{P}{n_t}HH^T\right)\right] \\ &= \sum_{i=1}^{n_{\min}} \mathbb{E}\left[C_{AWGN}\left(\frac{P}{n_t}\lambda_i^2\right)\right], \end{aligned}$$

where $C_{AWGN}(P) = \frac{1}{2}log(1+P)$ is the capacity of the AWGN with P being the power and SNR (due to unit noise variance). $n_{min} = min(n_r, n_t)$, λ_i^2 $i \in \{1, \cdots, n_{min}\}$ are eigenvalues of $\mathbf{HH}^H$.

Channel Variance

Assume that $\mathbb{P}[\text{rank}\mathbf{H} > 1] > 0$, then $V(P) = V_{iid}(P)$,

$$\begin{aligned} V_{iid}(P) = T\text{Var}\left(\sum_{i=1}^{n_{\min}} C_{AWGN}\left(\frac{P}{n_t}\lambda_i^2\right)\right) \\ + \sum_{i=1}^{n_{\min}} \mathbb{E}\left[V_{AWGN}\left(\frac{P}{n_t}\lambda_i^2\right)\right] \\ + \left(\frac{P}{n_t}\right)^2\left(\eta_1 - \frac{\eta_2}{n_t}\right) \end{aligned}$$

λ_i^2 $i \in \{1, \cdots, n_{min}\}$ are eigenvalues of $\mathbf{HH}^H$.

$$V_{AWGN}(P) = \frac{log^2 e}{2}(1 - \frac{1}{(1+P)^2})$$

$$\begin{aligned} c(\sigma) &\triangleq \frac{\sigma}{1 + \frac{P}{n_t}\sigma} \\ \eta_1 &\triangleq \frac{\log^2 e}{2}\sum_{i=1}^{n_{\min}} \mathbb{E}\left[c^2(\lambda_i^2)\right] \\ \eta_2 &\triangleq \frac{\log^2 e}{2}\left(\sum_{i=1}^{n_{\min}} \mathbb{E}\left[c(\lambda_i^2)\right]\right)^2 \end{aligned}$$

Obtaining a closed form

Under high SNR $P \to \infty$. Let $n_{min} = min(n_t, n_r)$, $\alpha = P/n_t$ and λ_i the eigenvalues of $\mathbf{HH}^H$.

If we assume P is a large constant ($P \to \infty$),

$$\text{Var}\left[\sum_{i=1}^{n_{\min}} \ln(1 + \alpha\lambda_i)\right] \sim \text{Var}\left[\sum_{i=1}^{n_{\min}} \ln\lambda_i\right]$$

$$V_{\text{AWGN}}(\alpha\lambda_i) = (\ln e)^2 \frac{(1+\alpha\lambda_i)^2}{\alpha\lambda_i\,(\alpha\lambda_i + 2)} \sim (\ln e)^2, \quad \text{as } \alpha \to \infty$$

$$\text{Var}\left[\sum_{i=1}^{n_{\min}} C_{\text{AWGN}}(\alpha\lambda_i)\right] \sim \text{Var}\left[\sum_{i=1}^{n_{\min}} \ln(\alpha\lambda_i)\right] = \text{Var}\left[n_{\min}\ln\alpha + \sum_{i=1}^{n_{\min}} \ln\lambda_i\right] = \text{Var}\left[\sum_{i=1}^{n_{\min}} \ln\lambda_i\right]$$

$$c(\sigma) \triangleq \frac{\sigma}{1 + \frac{P}{n_t}\sigma}, \quad \alpha \triangleq \frac{P}{n_t}.$$

Step 1: High-SNR expansion of $c(\sigma)$

$$c(\sigma) = \frac{\sigma}{1 + \alpha\sigma} = \frac{1}{\alpha} \cdot \frac{1}{1 + \frac{1}{\alpha\sigma}}.$$

Using the series $\frac{1}{1+x} = 1 - x + x^2 + \mathcal{O}(x^3)$ with $x = \frac{1}{\alpha\sigma}$:

$$c(\sigma) = \frac{1}{\alpha} - \frac{1}{\alpha^2\sigma} + \frac{1}{\alpha^3\sigma^2} + \mathcal{O}\left(\frac{1}{\alpha^4}\right).$$

—

Step 2: Expansion of η_1

$$c(\sigma)^2 = \frac{1}{\alpha^2} - \frac{2}{\alpha^3\sigma} + \mathcal{O}\left(\frac{1}{\alpha^4}\right).$$

$$\eta_1 = \frac{\log^2 e}{2} \sum_{i=1}^{n_{\min}} \mathbb{E}[c(\lambda_i^2)^2] = \frac{\log^2 e}{2} \left(\frac{n_{\min}}{\alpha^2} - \frac{2}{\alpha^3} \sum_{i=1}^{n_{\min}} \mathbb{E}\left[\tfrac{1}{\lambda_i^2}\right] + \mathcal{O}\left(\frac{1}{\alpha^4}\right) \right).$$

—

Step 3: Expansion of η_2

$$\mathbb{E}[c(\lambda_i^2)] = \frac{1}{\alpha} - \frac{1}{\alpha^2}\mathbb{E}\left[\tfrac{1}{\lambda_i^2}\right] + \mathcal{O}\left(\frac{1}{\alpha^3}\right).$$

$$\sum_{i=1}^{n_{\min}} \mathbb{E}[c(\lambda_i^2)] = \frac{n_{\min}}{\alpha} - \frac{1}{\alpha^2} \sum_{i=1}^{n_{\min}} \mathbb{E}\left[\tfrac{1}{\lambda_i^2}\right] + \mathcal{O}\left(\frac{1}{\alpha^3}\right).$$

$$\left(\sum_{i=1}^{n_{\min}} \mathbb{E}[c(\lambda_i^2)] \right)^2 = \frac{n_{\min}^2}{\alpha^2} - \frac{2n_{\min}}{\alpha^3} \sum_{i=1}^{n_{\min}} \mathbb{E}\left[\tfrac{1}{\lambda_i^2}\right] + \mathcal{O}\left(\frac{1}{\alpha^4}\right).$$

$$\eta_2 = \frac{\log^2 e}{2} \left(\frac{n_{\min}^2}{\alpha^2} - \frac{2n_{\min}}{\alpha^3} \sum_{i=1}^{n_{\min}} \mathbb{E}\left[\tfrac{1}{\lambda_i^2}\right] + \mathcal{O}\left(\frac{1}{\alpha^4}\right) \right).$$

—

Step 4: Finding differenve of η_1 and η_2

$$\eta_1 - \eta_2 = \frac{\log^2 e}{2} \left(\frac{n_{\min} - n_{\min}^2}{\alpha^2} + \frac{2(n_{\min} - 1)}{\alpha^3} \sum_{i=1}^{n_{\min}} \mathbb{E}\left[\tfrac{1}{\lambda_i^2}\right] + \mathcal{O}\left(\frac{1}{\alpha^4}\right) \right).$$

$$\frac{P}{n_t}(\eta_1 - \eta_2) = \alpha(\eta_1 - \eta_2) = \frac{\log^2 e}{2} \left(\frac{n_{\min} - n_{\min}^2}{\alpha} + \frac{2(n_{\min} - 1)}{\alpha^2} \sum_{i=1}^{n_{\min}} \mathbb{E}\left[\tfrac{1}{\lambda_i^2}\right] + \mathcal{O}\left(\frac{1}{\alpha^2}\right) \right).$$

If we assume that $\alpha << \alpha^2 \approx \infty$,

$$\frac{P}{n_t}(\eta_1 - \eta_2) = \frac{\log^2 e}{2} \left(\frac{n_{\min} - n_{\min}^2}{\alpha} + \mathcal{O}\left(\frac{1}{\alpha^2}\right) \right)$$

—

Step 5: High-SNR conclusion

$$\frac{P}{n_t}(\eta_1-\eta_2)=\frac{\log^2 e}{2}\left(\frac{n_{\min}-n_{\min}^2}{\alpha}+\mathcal{O}\left(\frac{1}{\alpha^2}\right)\right)\quad\longrightarrow\ 0,\quad \text{as } P\to\infty.$$

Therefore,

$$V(P)=V_{\text{iid}}(P)\sim T\,\mathrm{Var}\left[\sum_{i=1}^{n_{\min}}\ln\lambda_i\right]+n_{\min}(\ln e)^2+\frac{\log^2 e}{2}\left(\frac{n_{\min}-n_{\min}^2}{\alpha}\right)+\mathcal{O}\left(\frac{1}{\alpha^2}\right),\quad \alpha=\frac{P}{n_t}.$$

We can further simplify this by truly considering $\alpha\to\infty$, then the last terms become 0. However i have kept these terms in the equation to give role to the precoder $\mathbf{F}$ in our alternating optimization proccess.

We can further expand $T\,\mathrm{Var}[\sum_{i=1}^{n_{\min}}\ln\lambda_i^2]$ with the trigamma function $\psi 1(x)$

$$\mathrm{Var}\left[\sum_{i=1}^{n_{\min}}\ln\lambda_i^2\right]=\sum_{i=1}^{n_{\min}}\psi_1(n_{\max}-i+1),$$
$$n_{\min}=\min(n_t,n_r),\quad n_{\max}=\max(n_t,n_r),$$
$$\alpha=\frac{P}{n_t}$$

where $\psi_1(x)$ is,

$$\psi_1(x)=\sum_{k=0}^{\infty}\frac{1}{(x+k)^2}.$$

In the assumtption of very large x,

$$\psi_1(x)=\frac{1}{x}+\frac{1}{2x^2}+\frac{1}{6x^3}-\frac{1}{30x^5}+\frac{1}{42x^7}-\cdots$$

Thus when we consider $\sum_{i=1}^{n_{\min}}\psi_1(n_{\max}-i+1)$ to obtain a closed form for the trigamma function, we need to assume a very large $n_{\max}-i+1$, thus if $n_{\max}=\max(n_r,n_t)\to\infty$,

$$\psi_1(n_{\max}-i+1)=\frac{1}{n_{\max}-i+1}+\frac{1}{2(n_{\max}-i+1)^2}+\frac{1}{6(n_{\max}-i+1)^3}-\frac{1}{30(n_{\max}-i+1)^5+}+\cdots$$

or using the Bernoulli numbers B,

$$\psi_1(n_{\max}-i+1)=\sum_{k=0}^{\infty}\frac{B_{2k}}{(n_{\max}-i+1)^{2k+1}}$$

Thus we can reduce $\mathrm{Var}[\sum_{i=1}^{n_{\min}}\ln\lambda_i^2]$ to,

$$\mathrm{Var}\left[\sum_{i=1}^{n_{\min}}\ln\lambda_i^2\right]\approx\sum_{i=1}^{n_{\min}}\psi_1(n_{\max}-i+1)=\sum_{i=1}^{n_{\min}}\left[\frac{1}{n_{\max}-i+1}+\frac{1}{2(n_{\max}-i+1)^2}+\cdots\right.$$

Final closed form for Channel Variance

Thus the final $V(P)$ under the assumtion of, $P \to \infty$ and $n_{\max} \to \infty$ is,

$$V(P) \;=\; V_{\text{iid}}(P) \;\approx\; \sum_{i=1}^{n_{\min}} \left[\frac{1}{n_{\max} - i + 1} + \frac{1}{2(n_{\max} - i + 1)^2} + \cdots \right]$$

$$+\; n_{\min} (\ln e)^2$$

$$+\; \frac{\log^2 e}{2} \left(\frac{n_{\min} - n_{\min}^2}{\alpha} \right)$$

$$+\; \mathcal{O}\big(\tfrac{1}{\alpha^2}\big)\,, \qquad \alpha = \tfrac{P}{n_t}.$$

References

[1] B. Rimoldi and R. Urbanke, "A rate-splitting approach to the Gaussian multiple-access channel," *IEEE Trans. Inf. Theory*, vol. 42, no. 2, pp. 364–375, Mar. 1996.

[2] Z. Yang, M. Chen, W. Saad, W. Xu and M. Shikh-Bahaei, "Sum-rate maximization of uplink rate splitting multiple access (RSMA) communication," *Proc. of the IEEE Global Communications Conference (GLOBECOM)*, Waikoloa, HI, USA, pp. 1-6, 2019. doi: 10.1109/GLOBECOM38437.2019.9013344.

[3] G. Durisi, T. Koch and P. Popovski, "Toward massive, ultrareliable, and low-latency wireless communication with short packets," *Proceedings of the IEEE*, vol. 104(9), pp. 1711-1726, 2016.

[4] X. You *et al.*, "Closed-form approximation for performance bound of finite blocklength massive MIMO transmission," *IEEE Trans. on Communications*, vol. 71, no. 12, pp. 6939-6951, Dec. 2023

[5] Y. Zhang, W. Cheng, J. Wang and W. Zhang, "Performance analysis and blocklength minimization of uplink RSMA for short packet transmissions in URLLC," *Proc. of the IEEE Global Communications Conference (GLOBECOM)*, Kuala Lumpur, Malaysia, pp. 6765-6770, 2023.

[6] Q. Shi, M. Razaviyayn, Z.-Q. Luo, and C. He, "An iteratively weighted MMSE approach to distributed sum-utility maximization for a MIMO interfering broadcast channel," *IEEE Trans. on Signal Processing*, vol. 59(9), pp. 4331–4340, Sep. 2011.

[7] A. Collins and Y. Polyanskiy, "Coherent Multiple-Antenna Block-Fading Channels at Finite Blocklength," in IEEE Transactions on Information Theory, vol. 65, no. 1, pp. 380-405, Jan. 2019